

Machine Learning 2026
Assignment 1 (Theory)

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Q1.) Grouped (Binomial) vs. Individual (Bernoulli) logistic regression.

(a) log likelihood $\ell(w)$ for a dataset of M groups, where group k has n_k trials and y_k successes
 $p_k = \sigma(w^T x_k)$

for group k - $y_k \sim \text{Binomial}(n_k, p_k)$

$$p_k = \sigma(w^T x_k) \text{ so}$$

$$P(y_k | x_k, w) = \binom{n_k}{y_k} p_k^{y_k} (1-p_k)^{n_k-y_k}$$

(Assuming groups are independent)

log likelihood

$$\ell(w) = \sum_{k=1}^M \left[\log \binom{n_k}{y_k} + y_k \log p_k + (n_k - y_k) \log(1-p_k) \right]$$

$$\text{using } \log p_k - \log(1-p_k) = w^T x_k \text{ and } \log(1-p_k) = -\log(1 + e^{w^T x_k})$$

$$\ell(w) = \sum_{k=1}^M [y_k w^T x_k - n_k \log(1 + e^{w^T x_k})] + \text{const}$$

(b) Equivalence with ~~log~~ Bernoulli likelihood.

every individual in group k has the same probability $p_k = \sigma(w^T x_k)$.

ungrouped log-likelihood

$$\ell_{\text{Bern}}(w) = \sum_{k=1}^M \sum_{i=1}^{n_k} [y_{ki} \log p_k + (1-y_{ki}) \log(1-p_k)]$$

Since p_k does not depend on i ,

$$\sum_{i=1}^{n_k} y_{ki} = y_k$$

$$\sum_{i=1}^{n_k} (1 - y_{ki}) = n_k - y_k$$

$$\Rightarrow \ell_{\text{Bern}}(w) = \sum_{k=1}^M [y_k \log p_k + (n_k - y_k) \log (1 - p_k)]$$

$$\Rightarrow \ell_{\text{Binom}}(w) = \ell_{\text{Bern}}(w) + \underbrace{\sum_{k=1}^M \log \binom{n_k}{y_k}}_{\text{independent of } w}$$

Since adding a constant doesn't change where the maximisation is so,

$$\hat{w}_{\text{Binom}} = \arg \max \ell_{\text{Binom}} = \arg \max \ell_{\text{Bern}} = \hat{w}_{\text{Bern}}$$

(checking via derivatives).

$$\sigma'(z) = \sigma(z)(1 - \sigma(z))$$

both have the same gradient & Hessian

$$\nabla \ell = \sum_k (y_k - n_k p_k) x_k$$

$$\nabla^2 \ell = - \sum_k n_k p_k (1 - p_k) x_k x_k^T \preceq 0$$

both have identical stationary conditions.

$$\sum_k (y_k - n_k p_k) x_k = 0$$

↳ share the same maximiser &

standard error of $\hat{w}$ matches too.

This tells us that grouping is lossless when every individual in a group has exactly the same x .

In that case, the counts (n_k, y_k) are sufficient statistics for w .

→ The order & identity of individuals within a group carry no information, since likelihood depends on them only through y_k .

Grouping just compresses the data → saves computation
→ no change to $\hat{w}$.

So grouping only changes $\hat{w}$ if it changes the information the likelihood sees.

~~If everyone in a group shares the same x~~

Grouping changes the estimate when individuals in a group have different covariates e.g.

binning continuous age into age groups or averaging x within a group.

Q2). $D = \{(x_i, y_i)\}_{i=1}^N$ $y_i \in \{0, 1\}$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$P(y_i = 1 | x_i, w) = \sigma(w^T x_i) = \frac{1}{1+e^{-w^T x_i}}$$

(1) let $\hat{p}_i = \sigma(w^T x_i)$.

each label is Bernoulli, so

$$P(y_i | x_i, w) = \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}$$

Since the data are iid the joint conditional likelihood is:

$$P(Y|X, w) = \prod_{i=1}^N \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}$$

taking logs

$$\log P(Y|X, w) = \sum_{i=1}^N [y_i \log \hat{p}_i + (1-y_i) \log (1 - \hat{p}_i)]$$

$$J_{MLE}(w) = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{p}_i + (1-y_i) \log (1 - \hat{p}_i)]$$

$$= -\frac{1}{N} \log P(Y|X, w)$$

Since log is strictly increasing, maximising $P(Y|X, w)$ is the same as maximising $\log P(Y|X, w)$.

multiplying by $-\frac{1}{N} < 0$ turns the maximisation into a minimisation, w/o moving the optimum.

$$\hat{w}_{MLE} = \arg \max P(Y|X, w) = \arg \min J_{MLE}(w)$$

(b) By Bayes' Rule,

$$P(w|X, Y) = \frac{P(Y|X, w) P(w)}{P(Y|X)}$$

denominator doesn't depend on w , so

$$\log P(w|X, Y) = \log P(Y|X, w) + \log P(w) + \text{const.}$$

for $w \sim N(0, \sigma_0^2 I)$ in $\mathbb{R}^d$,

$$\log P(w) = -\frac{d}{2} \log(2\pi\sigma_0^2) - \frac{\|w\|_2^2}{2\sigma_0^2}$$

substituting $\frac{1}{N} \log P(Y|X, w)$.

$$P(Y|X, w) = -N J_{MLE}(w)$$

$$\log P(w|X, Y) = -N J_{MLE}(w) - \frac{\|w\|_2^2}{2\sigma_0^2} + \text{const.}$$

maximising this is equivalent to minimising its negative divided by $N > 0$.

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[J_{MLE}(w) + \frac{1}{2N\sigma_0^2} \|w\|_2^2 \right]$$

This is $J_{MAP}(w) = J_{MLE}(w) + \lambda \|w\|_2^2$ with

(If J_{MLE} is defined as summed rather than avg loss)

$$\lambda = \frac{1}{2\sigma_0^2}$$

$$\lambda = \frac{1}{2N\sigma_0^2}$$

A smaller prior variance σ_0^2 means a stronger prior belief that w is near 0, hence a larger λ .

(c).

from (b) $\lambda = \frac{1}{2N\sigma_0^2} \rightarrow 0$ as $N \rightarrow \infty$

equivalently,

$$\log P(w|x, N) = \underbrace{\sum_{i=1}^N \log P(y_i | x_i, w)}_{O(N)} - \underbrace{\frac{\|w\|_2^2}{2\sigma_0^2}}_{O(1)} + \text{const}$$

The likelihood term grows linearly w/ N while the prior term stays fixed, so the likelihood dominates.

@ optimum, $\hat{w}_{MAP}$ satisfies

$$\nabla J_{MLE}(\hat{w}_{MAP}) + 2\lambda \hat{w}_{MAP} = 0$$

As $\lambda \rightarrow 0$, $\nabla J_{MLE}(w) \Rightarrow 0$.

Since J_{MLE} is strictly convex, minimum coincide in limit, $\|\hat{w}_{MAP} - \hat{w}_{MLE}\|$ shrinks as $O(1/N)$

Both estimators are consistent.

by law of large numbers J_{MLE} converges to the expected loss, minimised at w^* , so $\hat{w}_{MAP}, \hat{w}_{MLE} \rightarrow w^*$.